

CS 550 Operating Systems, Spring 2026
Homework Assignment 1 (HW1)

Out: 2/11/2026 Wed.
Due: 2/18/2026 Wed. 23:59:59

Full points: 100

Note:

- There are six questions, each on a separate page.
- Read the submission instructions on the last page carefully and prepare the submission accordingly.

1. (18 points) For the code listed below, suppose each “printf” prints immediately when it executes, how many possible output of the program are there? List all the possible output.

```
int main() {  
    printf("0");  
  
    pid_t pid = fork();  
    if (pid == 0) {  
        printf("A");  
        printf("a");  
        _exit(0);  
    } else {  
        printf("B");  
        printf("b");  
        wait(NULL);  
    }  
  
    return 0;  
}
```

0AaBb
0ABab
0ABba
0BAab
0BbAa
0BAba

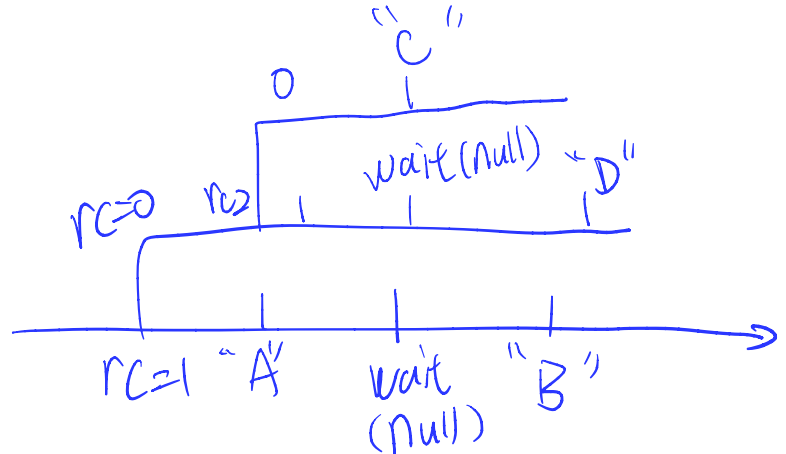
2. (18 points) In the following program, the executable program “/bin/prog” does not print anything, and `printf()` always prints immediately (i.e., there is no I/O buffering).

```
int main() {
    int rc = fork(); // fork-1

    if (rc > 0)
    {
        printf("A");
        wait(NULL); // wait-1
        printf("B");
    }
    else if (rc == 0)
    {
        int rc2 = fork(); // fork-2

        if (rc2 == 0)
        {
            printf("C");
        }
        else if (rc2 > 0)
        {
            wait(NULL); // wait-2
            printf("D");
        }
    }
    else
    {
        printf("E");
    }

    return 0;
}
```



- (1) If all the function invocations succeed, list all the possible outputs of the above program.
- (2) If the second invocation of `fork()` (i.e., fork-2) fails, list all the possible outputs of the above program.
- (3) If the second invocation of `wait()` (i.e., wait-2) fails, list all the possible outputs of the above program.

① ACDB
CADB
CDAB

② AB
AB.

③ ACDB CDAB
ADCB PACB
CADB DCAB

3. (20 points) In the following program, the executable program “/bin/prog” does not print anything, and `printf()` always prints immediately (i.e., there is no I/O buffering).

```
int main(int argc, char *argv[]) { // prog-2
    int rc = fork();

    if (rc == 0)
    {
        char *argv[] = { "/bin/prog", NULL };
        execv(argv[0], argv);
        printf("1");
    }
    else if (rc > 0)
    {
        printf("2");
    }

    printf("3");
    return 0;
}
```

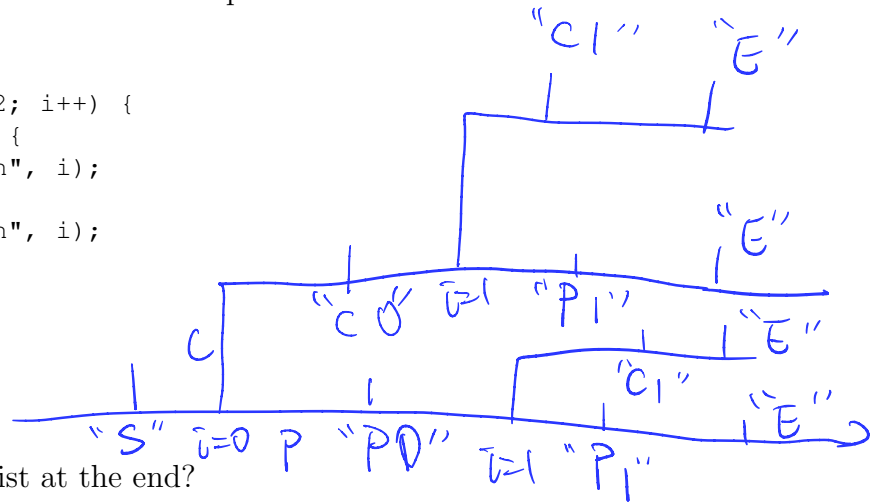
- (1) If the calls to `fork()` and `execv()` always succeed, list all the possible outputs of the above program.
- (2) If the calls to `fork()` and `execv()` might succeed, list all the possible outputs of the above program (i.e., prog-2) and explain how each of the outputs may happen.

(1) 2 3

(2) 3
1 3 2 3
2 3 1 3
2 1 3 3
1 2 3 3
2 3

4. (24 points) Read the code below and answer the questions after.

```
int main() {
    printf("S\n");
    for (int i = 0; i < 2; i++) {
        if (fork() == 0) {
            printf("C%d\n", i);
        } else {
            printf("P%d\n", i);
        }
    }
    printf("E\n");
    return 0;
}
```



(1) How many total processes exist at the end?

(2) How many times does 'S' print?

(3) How many times does 'E' print?

(4) How many times does 'C0' print?

(5) How many times does 'P1' print?

(6) How many times does 'P2' print?

(1) 4

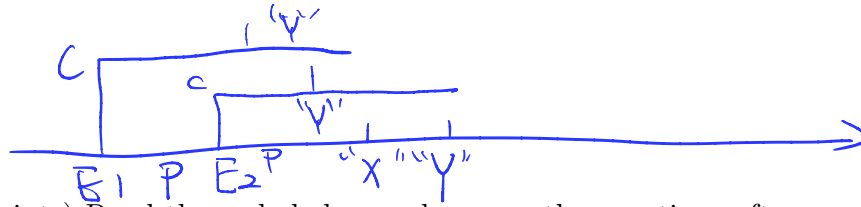
(2) 1

(3) 4

(4) 1

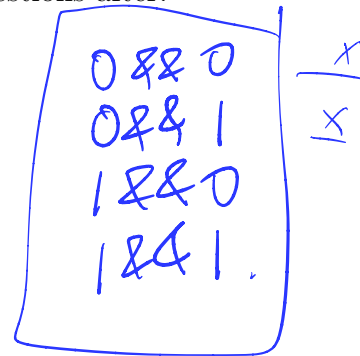
(5) 2

(6) 0



5. (20 points) Read the code below and answer the questions after.

```
int main() {
    if (fork() && fork())
        printf("X");
    printf("Y");
    return 0;
}
```



Here,

- Expression “E1 && E2” evaluates E1 first.
- If E1 evaluates to 0 (false), then E2 is not evaluated at all, and the entire expression evaluates to false.
- If E1 evaluates to nonzero (true), then E2 is evaluated, and the result of the entire expression is true only if both E1 and E2 are nonzero.

- (1) How many times does ‘X’ print? (1) 1
- (2) How many times does ‘Y’ print? (2) 3

Submission instructions

1. Type your answers using the text editor you like. Remember to include the index number of each question.
2. Export the file to PDF format.
3. Name the PDF file based on your BU email ID. For example, if your BU email is “abc@binghamton.edu”, then the PDF file should be named as “hw1-abc.pdf”.
4. Failure to follow the above instructions will lead to 5 points off.
5. Submit the PDF file to Brightspace before the deadline.

5. (20 points) Read the code below and answer the questions after.

```
int main() {  
    if (fork() && fork())  
        printf("X");  
    printf("Y");  
    return 0;  
}
```

Here,

- Expression “E1 && E2” evaluates E1 first.
- If E1 evaluates to 0 (false), then E2 is *not evaluated at all*, and the entire expression evaluates to false.
- If E1 evaluates to nonzero (true), then E2 is evaluated, and the result of the entire expression is true only if both E1 and E2 are nonzero.

(1) How many times does ‘X’ print?

(2) How many times does ‘Y’ print?

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